

Learning how to Prove Workshop Week 7

Groups



Cyclic groups

1. Consider the group $G \times H$, where $G = \langle g \rangle$ and $H = \langle h \rangle$ are cyclic groups with $|G| = o(g) = m$ and $|H| = o(h) = n$. Suppose that $\gcd(m, n) = 1$.
 - (a) Prove that the order of the element $(g, h) \in G \times H$ is mn ; that is, $o(g, h) = mn$.
 - (b) Conclude that $G \times H$ is cyclic.

Solution: (a) We have that

$$(g, h)^{mn} = (g^{mn}, h^{mn}) = ((g^m)^n, (h^n)^m) = (1, 1).$$

Moreover, if $(g, h)^k = (1, 1)$, then $g^k = 1$ and $h^k = 1$ and so $m|k$ and $n|k$, but since $\gcd(m, n) = 1$ we also have that $mn|k$. Thus $o(g, h) = mn$, as desired.

(b) Since $|G \times H| = mn$ and $o(g, h) = mn$ by (a), we have that $G \times H = \langle (g, h) \rangle$.

2. Let $G = \langle g \rangle$ be a cyclic group of order n . Consider $A = \langle g^a \rangle$ and $B = \langle g^b \rangle$ two subgroups of G . Prove that $A \subseteq B$ if and only if $a \equiv qb \pmod{n}$ for some $q \in \mathbb{Z}$.

Solution: Suppose first that $A \subseteq B$. Then we have that $g^a \in B$, and so

$$g^a = (g^b)^q = g^{qb}, \text{ for some integer } q.$$

But then $a \equiv qb \pmod{n}$, by applying the theorem that describes the cyclic subgroup generated by an element (see Theorem 3.25 (i) (b)). Conversely, assume that $a \equiv qb \pmod{n}$ for some $q \in \mathbb{Z}$. To prove that $A \subseteq B$ it is enough to show that $g^a \in B$. From $a \equiv qb \pmod{n}$ we obtain that $a - qb = rn$ for some integer r . Using the *exponent laws* and the fact that the order of g is n , we have that

$$g^{a-qb} = g^{rn} = (g^n)^r = 1^r = 1,$$

which implies that

$$1 = g^{a-qb} = g^a g^{-qb} = g^a (g^{qb})^{-1} \Rightarrow g^a = g^{qb} = (g^b)^q \in B,$$

as desired.